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Major Task – Pneumatic Press Machine

MDP 494: Advanced Manufacturing Technology and Prototyping

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Date:

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Major Task – Pneumatic Press Machine

Small pneumatic press machines are compact, air-powered devices designed for light to medium-duty industrial applications. They are generally benchtop-sized and are known for their ease of use and low maintenance requirements.



Main Industrial Applications:

Assembly: Press-fitting bearings, bushings, or pins into housings.

Forming and Cutting: Punching, stamping, bending, or forming metal sheets and plastic materials using dies.

Joining: Operations such as riveting, ...etc.

Marking: Applying logos, serial numbers, or other markings to components.

Design Data:

Pressing Force = ~5 tons

Operating Pressure = 0.4–0.8 MPa (compressed air supply).

Stroke Length = 50–150 mm

Control Type = Foot pedal or dual-hand controls.

Goals of the project:

- 1 To prepare students to work in teams to solve manufacturing and prototyping problems.
- 2 To develop the necessary skills for using computer simulation and integrated manufacturing in an industrial environment.
- 3 To detect any engineering problems early in the design process, and to learn how to bring products to market faster and with a lower design cost.

General Requirements to be considered during the design and manufacture of the Pneumatic Press Machine:

1. **Versatility:** Versatility in terms of the materials it can handle and the types of operations it can perform.
2. **Precision and Consistency:** Accuracy in applying pressure or force is vital.
3. **Durability and Reliability:** A Pneumatic Press Machine that is robust and durable, capable of withstanding repeated use without compromising its performance.
4. **Ease of Use:** Ergonomic features that reduce fatigue during extended use.

5. Customizability: A Pneumatic Press Machine that can be customized to suit specific needs.

6. Safety: The Pneumatic Press Machine is to be designed with safety mechanisms to prevent injuries during operation.

7. Cost-Effectiveness: The Pneumatic Press Machine should consider a compromise for a balance between price and the features offered.

Specific Requirements:

1. Choose suitable materials for different machine parts cut to real-life applications. Using the proper materials will increase operating efficiency, improve productivity and help reduce necessary maintenance.
2. The use of a safety system is strongly recommended to reduce potential health hazards.
3. A mechanism for holding the job is necessary.
4. Define the different machine parts with a Complete Exploded View.

Steps for Performing the Project:

- 1- Prepare all necessary technical documentation starting with hand sketches and ending with fully dimensioned CAD files. (Search for actual dimensions and specifications).
- 2- Fabricate a physical fully functioning prototype to verify form, fit, and function (overall height of the prototype around 300 mm).
- 3- Manufacture the product using Non-Conventional Machining Processes and AM techniques.
- 4- Test and analyze the functional prototype for design and functional flaws.
- 5- Prepare the necessary documentation including marketing illustrations and catalog data.
- 6- Prepare production documentation for the actual fabrication of the product, including selection of processes, material selection and any other items you find necessary.
- 7- Present your project in a fully detailed report.

Required Skills:

1. Basic knowledge and practical skills of manufacturing technology and prototyping.
2. Basic knowledge of design and stress analysis.
3. Basic knowledge of materials.
4. Practical skills to design and build products using additive manufacturing.
5. Technical writing.
6. Presentation skills.
7. Self-learning.